



Inspiring Excellence

Network Layer: IPv4 Static Routing

Lecture 11 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Objectives

Static Routing

- Standard static routing
 - Directly Attached / Connected
 - Next Hop / Recursive
- Default Routing
- Floating Static Routing

Learning About Networks

A router can learn about remote networks in one of two ways:

- **Manually** - Remote networks are manually entered into the route table using static routes.
- **Dynamically** - Remote networks are automatically learned using a dynamic routing protocol.

Learning About Networks

Table of R1

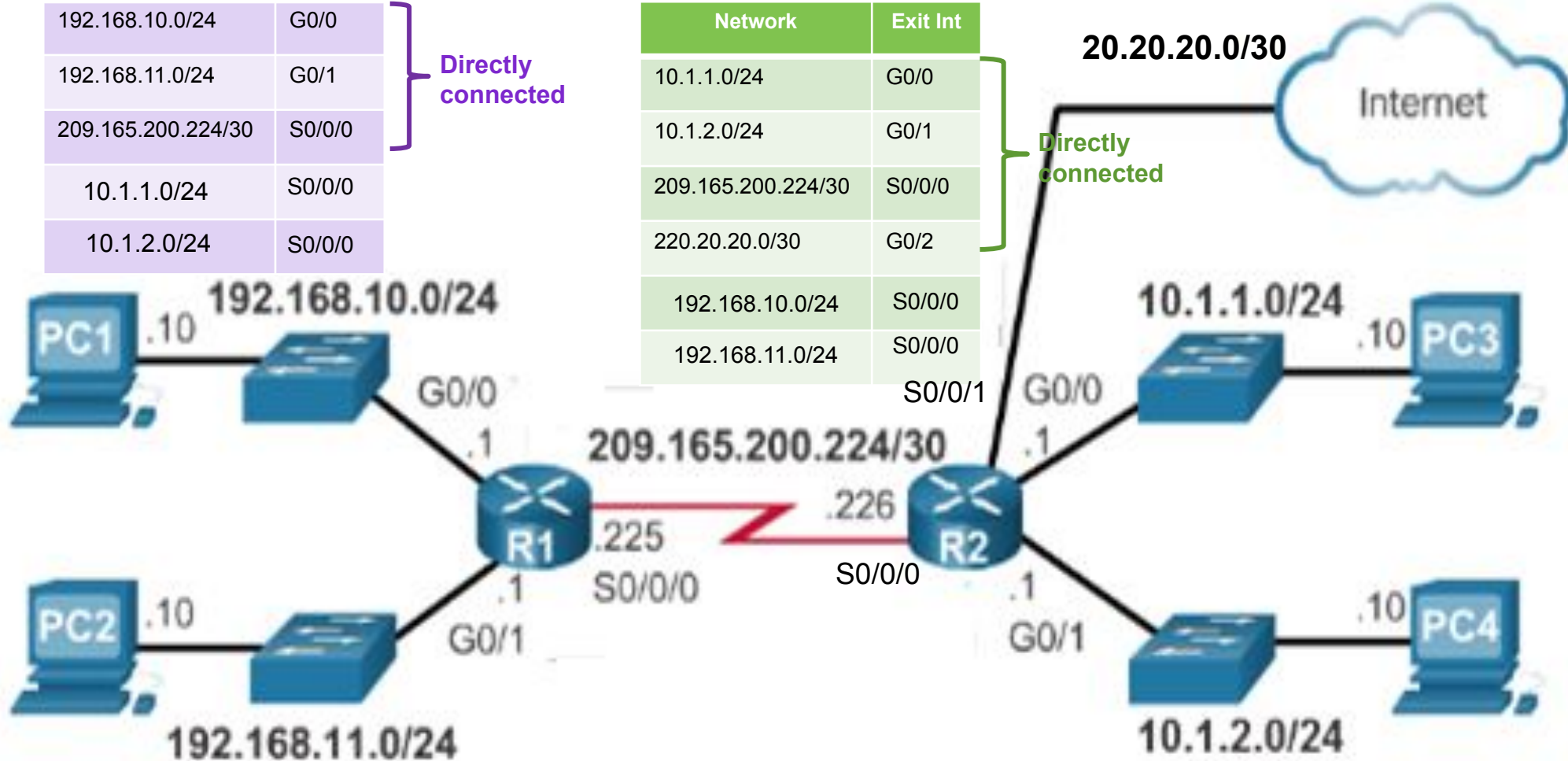
Network	Exit Int
192.168.10.0/24	G0/0
192.168.11.0/24	G0/1
209.165.200.224/30	S0/0/0
10.1.1.0/24	S0/0/0
10.1.2.0/24	S0/0/0

Directly
connected

Table of R2

Network	Exit Int
10.1.1.0/24	G0/0
10.1.2.0/24	G0/1
209.165.200.224/30	S0/0/0
220.20.20.0/30	G0/2
192.168.10.0/24	S0/0/0
192.168.11.0/24	S0/0/0

Directly
connected



Static Route

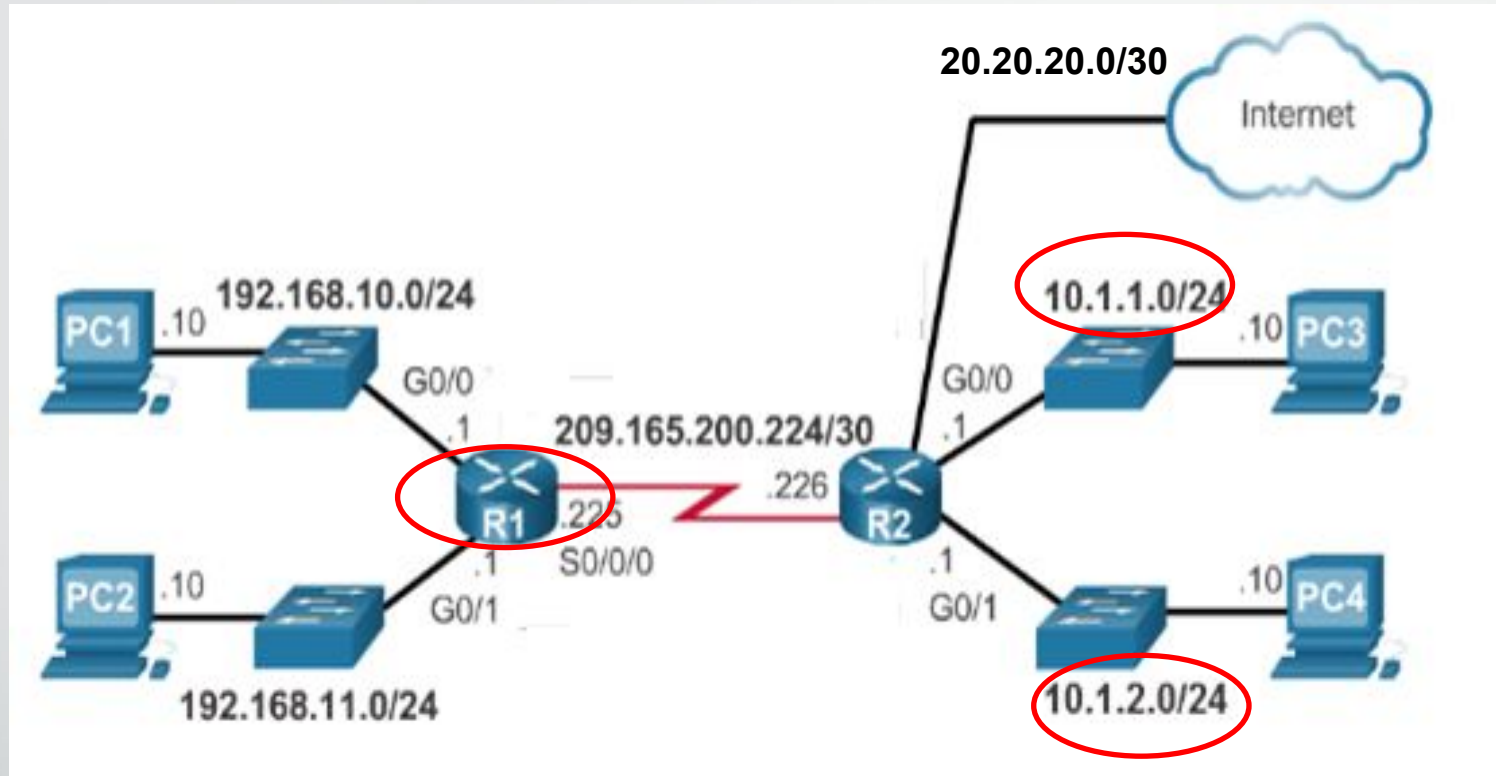
- Manually configured route
- Used to reach networks not directly connected
- Does not update automatically
- Changes must be applied manually

Static Route Types

Static Routes are often used to:

- Reach a **specific remote network**
- Set a **default route** (gateway of last resort)
- Provide **a backup path** (floating static route)
- Reduce routing table entries (**summarization**)

Static Route : How??



R1(config)# command Parameter 1 Parameter 2 Parameter 3

ip route The remote network address Subnet Mask Exit interface or next hop IP address

R1(config)# ip route 10.1.1.0 255.255.255.0 S0/0/0 209.165.200.226

Routing Table of R1 after the static route command

Using Exit interface

```
R1# show ip route
```

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
```

```
C 192.168.11.0/24 is directly connected, GigabitEthernet0/1
```

```
C 209.165.200.224/30 is directly connected, Serial0/0/0
```

```
S 10.1.1.0/24 is directly connected, Serial0/0/0
```

Using next hop IP address

```
R1# show ip route
```

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
```

```
C 192.168.11.0/24 is directly connected, GigabitEthernet0/1
```

```
C 209.165.200.224/30 is directly connected, Serial0/0/0
```

```
S 10.1.1.0/24 [1/0] via 209.165.200.226
```


Recursive Lookup

R1# show ip route

C 192.168.10.0/24 is directly connected, GigabitEthernet0/0

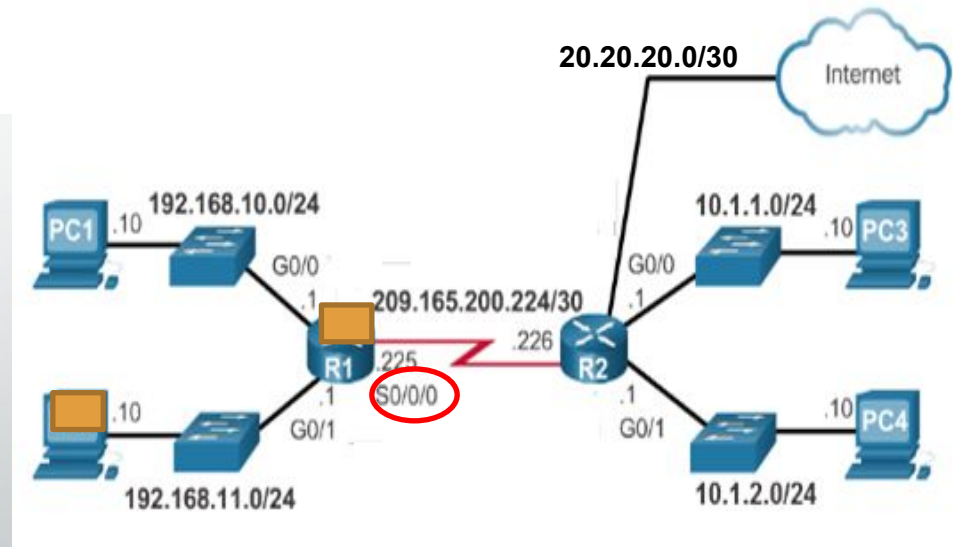
~~C 192.168.11.0/24 is directly connected, GigabitEthernet0/1~~

~~C 209.165.200.224/30 is directly connected, Serial0/0/0~~

S 10.1.1.0/24 [1/0] via 209.165.200.226

Router R1 receives a packet which has the destination address is **10.1.1.24**

How **many times** did the Router R1 look up the routing table?



Routing Table of R1 after the static route command

Using next hop IP address

R1# show ip route

C 192.168.10.0/24 is directly connected, GigabitEthernet0/0

C 192.168.11.0/24 is directly connected, GigabitEthernet0/1

C 209.165.200.224/30 is directly connected, Serial0/0/0

S 10.1.1.0/24 [1/0] via 209.165.200.226

Type of route:
S - Static

Destination
Network

Cost of Path

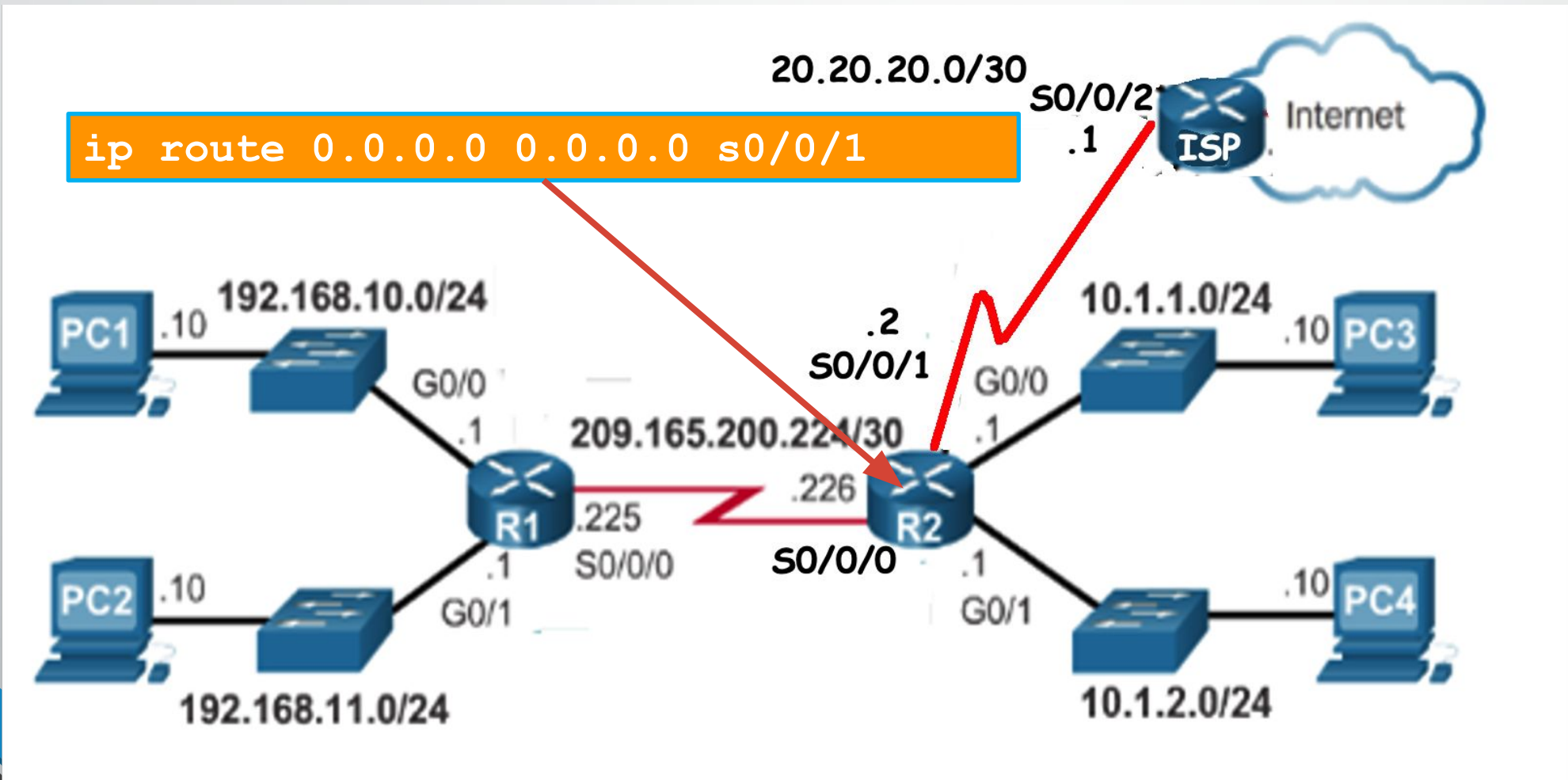
Next Hop IP
Or, Exit Interface

Administrative
Distance

AD : The trustworthiness of the source of the route

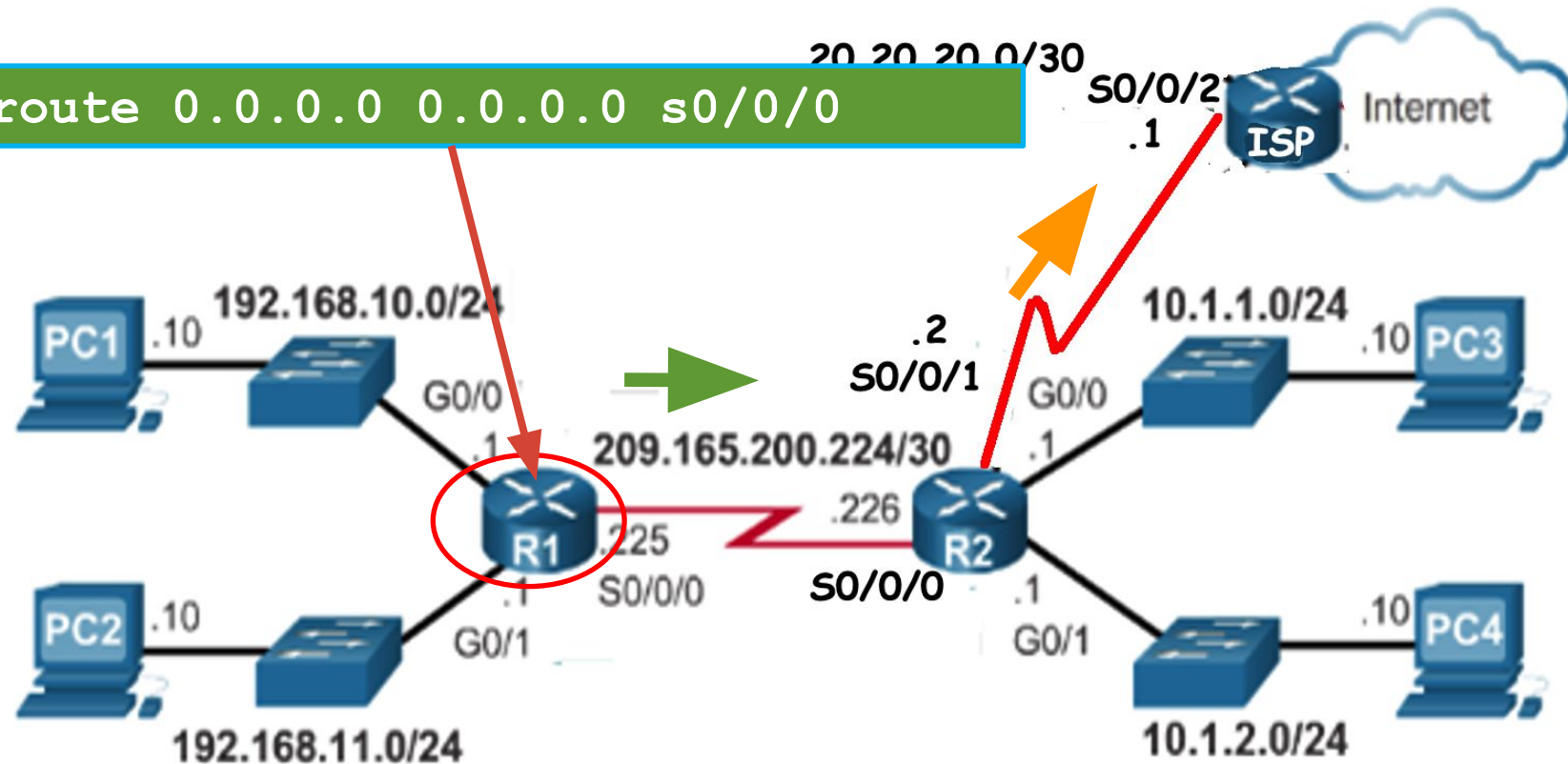
Default Route

All PCs of R1 and R2 need to go to the Internet



ip route Command

```
ip route 0.0.0.0 0.0.0.0 s0/0/0
```



What about R1?

R2(config)#	ip route	The remote network address	Subnet Mask	Exit interface or next hop IP address
	ip route	0.0.0.0	0.0.0.0	S0/0/1
		Any network		

Default Static Route in the Routing Table

```
R1# show ip route
```

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
```

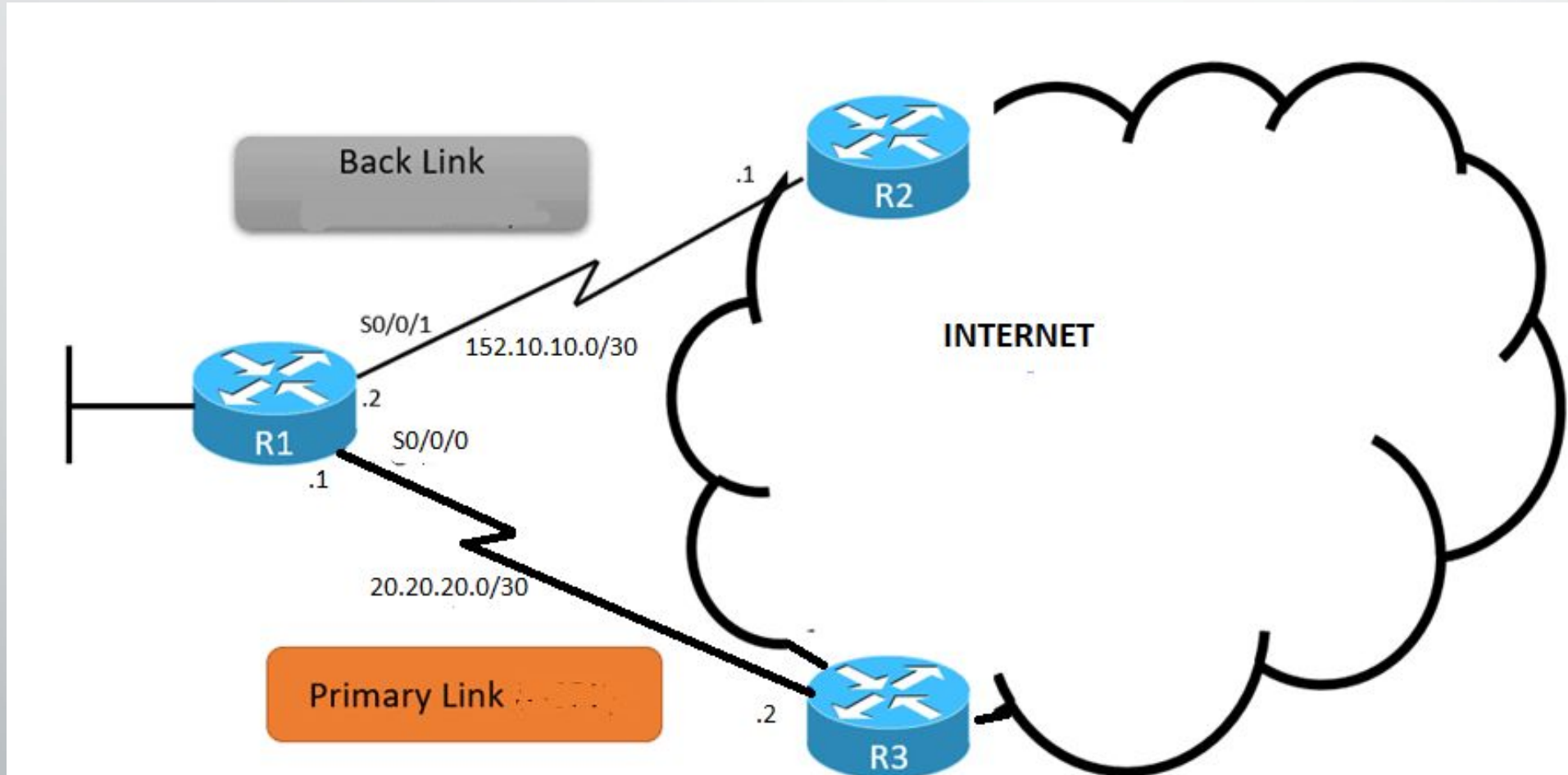
```
C 192.168.11.0/24 is directly connected, GigabitEthernet0/1
```

```
C 209.165.200.224/30 is directly connected, Serial0/0/0
```

```
S 10.1.1.0/24 [1/0] via 209.165.200.226
```

```
S* 0.0.0.0/0 is directly connected, Serial0/0/0
```

Floating Static Routing



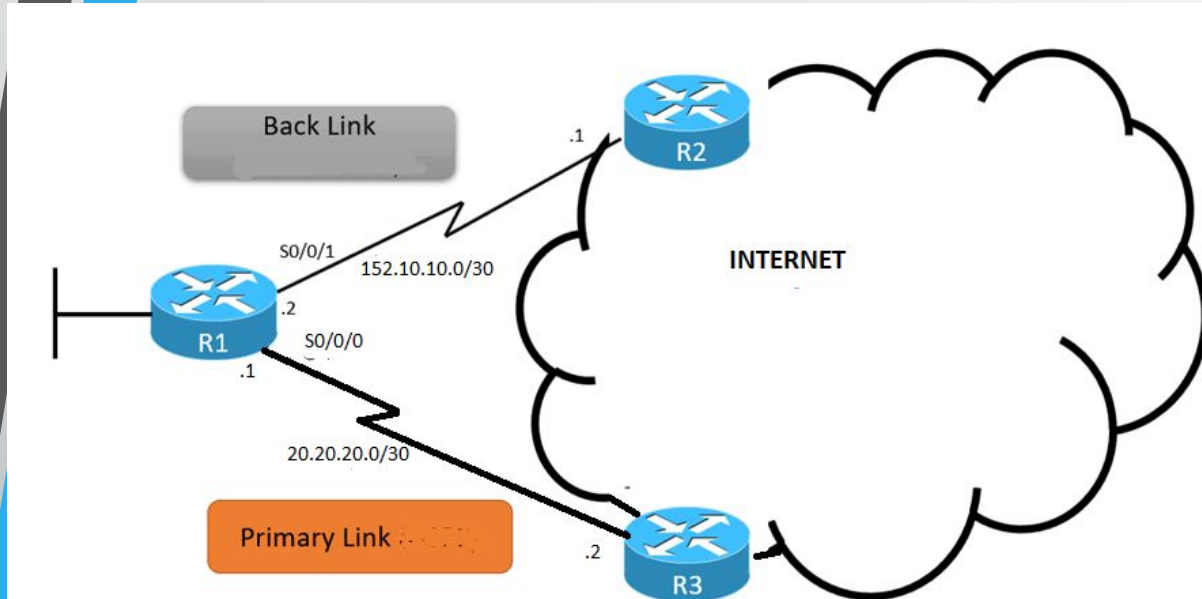
Source : <https://study-ccna.com/floating-static-route/>

Floating Default Static Routing

- R1 Routing Table (Partial)

S* 0.0.0.0/0 is directly connected, S0/0/0
S* 0.0.0.0/0 is directly connected, S0/0/1

- Both has same cost same AD.
Which one to use?



- R1 (config)#ip route 0.0.0.0 0.0.0.0 S0/0/0
R1 (config)#ip route 0.0.0.0 0.0.0.0 S0/0/1 5

Present in
Routing Table

AD > default
Value

Present in Router
Configuration File

Static Route Advantages

- Simple and predictable
- More secure (not advertised)
- Low CPU/memory usage

Static Route Disadvantages

- Time-consuming to configure
- Not scalable for large networks
- No automatic failover
- Requires full network knowledge

The End